

DA2 for DIGITAL IMAGE PROCESSING – FEB 2025

Instructions :

1. This is a team assignment. Maximum size of the team will be three only. Whether your team size is ONE OR TWO OR THREE – **ALL THREE PARTS NEED TO BE COMPLETED BY TEAM. FAILURE TO COMPLETE WILL RESULT IN LOSS OF MARKS OF SECTIONS NOT SUBMITTED. NO NEGOTIATIONS PERMITTED.**

2. DO not lend your assignment for copying. Excessive similarity between any two assignments will result in all the teams getting penalized. Please also note this validation will be done possibly after CAT-1 and handwriting similarity will be checked in case of doubt. **THE CONCERNED TEAM MAY BE ASKED TO SUBMIT HARD-COPY IN DOUBT.**

3. THE ASSIGNMENT SHOULD HAVE A SEPARATE TITLE PAGE WHICH SHOULD MENTION TEAM LEAD AND TEAM MEMBERS NAME AND REGISTER NUMBERS. FAILURE TO MENTION TEAM MEMBERS NAME will result in that decision that team member will NOT be part of the team. NO one can be added later by in person meeting/phone msg/ text msg /whatsapp msg or by any other means.

4. ONLY ONE SUBMISSION PER TEAM BY TEAM LEADER ONLY. Team members may check other's part before submission – All team members will GET SAME MARK and all of them are EQUALLY RESPONSIBLE FOR CORRECTNESS, NEATNESS AND SUBMISSION IN TIME. Blaming any person NOT PERMITTED.

WARNING: MULTIPLE SUBMISSIONS BY SAME TEAM WILL RESULT IN AUTOMATIC REDUCTION OF 0.5 MARKS FOR ALL MEMBERS OF TEAM. NO NEGOTIATIONS.

5. Answers to all questions [Except Q7] should be neatly written in hand, photographed and converted to pdf and uploaded. Typewritten responses NOT acceptable and will get zero.

6. For Q7, the input photograph, the code and the output, and the explanation could be provided. Code and explanation could be TYPEWRITTEN.

7. **LAST DATE OF SUBMISSION IS 12th March 2025 23:59 hours – Late submissions will result in some penalty proportional to the delay.**

Q1. [a] If $A = [a \ b \ c \ d]$ find 1-D 4 point DFT of signal A using matrix method.

[b] If $P = \{p, t+jv, q, t-jv\}$ where $j = \sqrt{-1}$ find 1-D 4-point IDFT of P using matrix method.

Q2. [a] A 4×4 image is given here. [a-p are integers representing intensity in the range 0-255].

Find 2-D DFT of the below image using matrix equation only.

A	b	c	d
E	f	g	h
I	J	k	l
M	n	o	p

Q2. [b] Above table represents 2-D DFT of an image.

[A-P represents integers in the range 0-255].

Perform 2-D IDFT to find the original image using matrix equation only.

A	B	C	D
E	F	G	H
I	I	K	L
M	N	O	P

Q3. The following Table 4.1 in the book 'Digital Image processing' by Gonzalez and Woods [fourth edition – Chapter 4] presents the properties of DFT and its inverse. **Prove all the properties.**

TABLE 4.1 SYMMETRY PROPERTIES OF 2-D DFT AND ITS INVERSE

TABLE 4.1
Some symmetry properties of the 2-D DFT and its inverse. $R(u, v)$ and $I(u, v)$ are the real and imaginary parts of $F(u, v)$, respectively. Use of the word *complex* indicates that a function has nonzero real and imaginary parts.

	Spatial Domain ¹	Frequency Domain ¹
1)	$f(x, y)$ real	$F^*(u, v) = F(-u, -v)$
2)	$f(x, y)$ imaginary	$F^*(-u, -v) = -F(u, v)$
3)	$f(x, y)$ real	$R(u, v)$ even; $I(u, v)$ odd
4)	$f(x, y)$ imaginary	$R(u, v)$ odd; $I(u, v)$ even
5)	$f(-x, -y)$ real	$F^*(u, v)$ complex
6)	$f(-x, -y)$ complex	$F(-u, -v)$ complex
7)	$f^*(x, y)$ complex	$F^*(-u, -v)$ complex
8)	$f(x, y)$ real and even	$F(u, v)$ real and even
9)	$f(x, y)$ real and odd	$F(u, v)$ imaginary and odd
10)	$f(x, y)$ imaginary and even	$F(u, v)$ imaginary and even
11)	$f(x, y)$ imaginary and odd	$F(u, v)$ real and odd
12)	$f(x, y)$ complex and even	$F(u, v)$ complex and even
13)	$f(x, y)$ complex and odd	$F(u, v)$ complex and odd

¹Recall that $x, y, u,$ and v are discrete (integer) variables, with x and u in the range $[0, M-1]$, and y and v in the range $[0, N-1]$. To say that a complex function is *even* means that its real and imaginary parts are even, and similarly for an *odd* complex function. As before, " $\Leftrightarrow$ " indicates a Fourier transform pair.

Q4. The following table 4.2 in the book ‘Digital Image Processing’ by ‘Gonzalez and Woods [fourth edition -Chapter 4] gives summary of DFT definitions and corresponding expressions.

PROVE Properties 7,8,11 and 12.

TABLE 4.3 SUMMARY OF 2-D DEFINITIONS AND CORRESPONDING EXPRESSIONS

TABLE 4.3
Summary of DFT
definitions and
corresponding
expressions.

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$
3) Spectrum	$ F(u, v) = [R^2(u, v) + I^2(u, v)]^{1/2} \quad R = \text{Real}(F); I = \text{Imag}(F)$
4) Phase angle	$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$
5) Polar representation	$F(u, v) = F(u, v) e^{j\phi(u, v)}$
6) Power spectrum	$P(u, v) = F(u, v) ^2$
7) Average value	$\bar{f} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1 M, v) = F(u, v + k_2 N)$ $= F(u + k_1, v + k_2 N)$ $f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N)$ $= f(x + k_1, y + k_2 N)$
9) Convolution	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$
10) Correlation	$(f \circledast h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n) h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12) Obtaining the IDFT using a DFT algorithm	$MN f^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MN f^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.

Q5. The following table in the book ‘Digital Image Processing’ by Gonzalez and Woods [fourth edition chapter 4] gives some properties of the 2-DFT and IDFT.

Prove properties 2,3,4,5,8,9,10,11.

The symbol ‘ $\Leftrightarrow$ ’ is a FT pair symbol. Proving this means you need to prove DFT in the forward direction and IDFT in the reverse direction. Both need to be proven If you prove only you get half the marks only.

TABLE 4.4 PROPERTIES OF DFT and IDFT

TABLE 4.4
Summary of
DFT pairs. The
closed-form
expressions in 12
and 13 are valid
only for
continuous
variables. They
can be used with
discrete variables
by sampling the
continuous
expressions.

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$a f_1(x, y) + b f_2(x, y) \Leftrightarrow a F_1(u, v) + b F_2(u, v)$
3) Translation (general)	$f(x, y) e^{j2\pi(u_0 x + v_0 y)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u, v) e^{-j2\pi(u x_0 + v y_0)}$
4) Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x, y) (-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u, v) (-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $r = \sqrt{x^2 + y^2} \quad \theta = \tan^{-1}(y/x) \quad \omega = \sqrt{u^2 + v^2} \quad \varphi = \tan^{-1}(v/u)$
6) Convolution theorem [†]	$f \star h(x, y) \Leftrightarrow (F \star H)(u, v)$ $(f \star h)(x, y) \Leftrightarrow (1/MN) [(F \star H)(u, v)]$
7) Correlation theorem [†]	$(f \circledast h)(x, y) \Leftrightarrow (F^* \star H)(u, v)$ $(f^* \circledast h)(x, y) \Leftrightarrow (1/MN) [(F^* \star H)(u, v)]$
8) Discrete unit impulse	$\delta(x, y) \Leftrightarrow 1$ $1 \Leftrightarrow MN \delta(u, v)$
9) Rectangle	$\text{rec}[a, b] \Leftrightarrow ab \frac{\sin(\pi u a)}{(\pi u a)} \frac{\sin(\pi v b)}{(\pi v b)} e^{-j\pi(u a + v b)}$
10) Sine	$\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow \frac{jMN}{2} [\delta(u + u_0, v + v_0) - \delta(u - u_0, v - v_0)]$
11) Cosine	$\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow \frac{1}{2} [\delta(u + u_0, v + v_0) + \delta(u - u_0, v - v_0)]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) Differentiation (the expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$).	$\left(\frac{\partial}{\partial t}\right)^n \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^n (j2\pi\nu)^n F(\mu, \nu)$ $\frac{\partial^n f(t, z)}{\partial t^n} \Leftrightarrow (j2\pi\mu)^n F(\mu, \nu); \quad \frac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$
13) Gaussian	$A 2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow A e^{-(\mu^2+\nu^2)/2\sigma^2} \quad (A \text{ is a constant})$

[†]Assumes that $f(x, y)$ and $h(x, y)$ have been properly padded. Convolution is associative, commutative, and distributive. Correlation is distributive (see Table 3.5). The products are elementwise products (see Section 2.6).

Q6. Study and explain in three pages the technique of ‘Fast Fourier Transform’ for computing DFT. Explain the technique with the help of ‘Butterfly diagram’ for 8 point 1-D DFT. Explain how much speedup happens in this technique.

Q7. Take your own greyscale photograph. [3 cm *3cm]

[a] Perform DFT using MATLAB or OpenCV.

[b] Scale up your photograph to 5cm*5cm and perform DFT again.

[c] Rotate the original photograph by 65 degrees and perform DFT by MATLAB or OpenCV

Explain in your own words what has happened to the DFT in [a],[b],[c].